

CHAPTER 7

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- 7.1. Let A be a non-Noetherian Ring and let Σ be the set of ideals in A which are not finitely generated. Show that Σ has maximal elements, and that the maximal elements of Σ are prime ideals. Hence, a ring in which every prime ideal is finitely generated is Noetherian (I. S. Cohen).

Proof.

□

- 7.3 Let $\mathfrak{a}$ be an irreducible ideal in a ring A . Then the following are equivalent:

- (a) $\mathfrak{a}$ is primary;
- (b) for every multiplicatively closed subset S of A , we have $(S^{-1}\mathfrak{a})^c = (\mathfrak{a} : x)$ for some $x \in S$;
- (c) the sequence $(\mathfrak{a} : x^n)$ is stationary, for every $x \in A$.

- 7.5. Let A be a Noetherian ring, B a finitely generated A -algebra, G a finite group of A -automorphism of B , and B^G the set of all elements of B which are left fixed by every element of G . Show that B^G is a finitely generated A -algebra.

- 7.6 If a finitely generated ring K is a field, it is a finite field.

- 7.8 If $A[x]$ is Noetherian, is A necessarily Noetherian.

- 7.9 Let A be a ring such that

- (a) for each maximal ideal $\mathfrak{m}$ of A , the local ring $A_{\mathfrak{m}}$ is Noetherian;
- (b) for each $x \neq 0$ in A , the set of maximal ideals of A which contain x is finite.

Show that A is Noetherian.

- 7.12 Let A be a ring and B a faithfully flat A -algebra (Ex. 2.6). If B is Noetherian, show that A is Noetherian.

- 7.13 Let $f : A \rightarrow B$ be a ring homomorphism of finite type and let $f^* : \text{Spec } B \rightarrow \text{Spec } A$ be the mapping associated with f . Show that the fibers of f^* are Noetherian subspaces of $\text{Spec } B$.

- 7.15. Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, and k its residue field, and let M be a finitely generated A -module. Then the following are equivalent:

- (a) M is free;
- (b) M is flat;
- (c) the mapping $\mathfrak{m} \otimes M \rightarrow A \otimes M$ is injective;
- (d) $\text{Tor}_1^A(k, M) = 0$.

- 7.16. Let A be a Noetherian local ring, and let M be a finitely generated A -module. Then the following are equivalent:

- (a) M is a flat A -module;
- (b) $M_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$ -module for all $\mathfrak{p} \in \text{Spec } A$;
- (c) $M_{\mathfrak{m}}$ is a free $A_{\mathfrak{m}}$ -module, for all maximal ideal $\mathfrak{m}$.

In other words, flat = locally free.

- 7.20 Let X be a topological space and let $\mathcal{F}$ be the smallest collection of subsets of X which contain all open subsets of X and is closed with respect to the formation of finite intersections and complements.
- (a) Show that a subset E of X belongs to $\mathcal{F}$ if and only if E is a finite union of sets of the form $U \cap C$, where U is open and C is closed.
 - (b) Suppose that X is irreducible and let $E \in \mathcal{F}$. Show that E is dense in X if and only if E contains a non-empty open set in X .
- 7.21 Let X be a Noetherian topological space and let $E \subset X$. Show that $E \in \mathcal{F}$ if and only if, for each irreducible closed set $X_0 \subset X$, either $\overline{E \cap X_0} \neq X_0$ or else $E \cap X_0$ contains a non-empty open subset of X_0 . The sets belonging to $\mathcal{F}$ are called the constructible subsets of X .
- 7.22 Let X be a Noetherian topological space and let E be a subset of X . Show that E is open in X if and only if, for each irreducible closed subset X_0 in X , either $E \cap X_0 = \emptyset$ or $E \cap X_0$ contains a non-empty open subset of X_0 .
- 7.23 Let A be a Noetherian ring, $f : A \rightarrow B$ a ring homomorphism of finite type (so that B is Noetherian). Let $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, and let $f^* : Y \rightarrow X$ be the mapping associated with f . Then the image under f^* of a constructible subset E of Y is a constructible subset of X .
- 7.24 With the notation and hypotheses of (Ex. 7.23), f^* is an open mapping if and only if f has the going down property.
- 7.25 Let A be Noetherian, $f : A \rightarrow B$ of finite type and flat. Then $f^* : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ is an open mapping.

Proof. By (Ex. 5.11), f has the going down property, whence f^* is an open mapping by (Ex. 7.24). $\square$